ASCEND TECHNICAL DOCUMENTATION

ASCEND 1.3.5

Software User and Technical Guide

DICOM ingestion, physical Lattice Radiotherapy analysis, spatial radiobiology research models, visualisation, validation, and export

Documentation revision	1.0
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Software identity	ASCEND 1.3.5
Repository baseline	Commit 65583e0 - ci/github-hosted-phase2
Primary audience	Research users, medical physicists, developers, and deployment staff

RESEARCH USE NOTICE

ASCEND is research and technical validation software. It is not approved clinical decision software. Layer 3.1 and Layer 3.2 results must not be used as treatment recommendations, toxicity estimates, or patient-specific outcome predictions.

Prepared from the current repository source, interface contracts, validation evidence, and automated cross-platform results.

Document purpose and control

This guide is the consolidated operating and technical reference for the current ASCEND repository. It explains what the software does, how to install and launch it, how to progress through the complete analysis workflow, how to interpret stored states, and where the validated scope ends.

Authoritative identity

The installed package and executable report ASCEND 1.3.5. This guide uses that identity. Later viewer-design records in the repository describe the unified spatial radiobiology interface now present in the same codebase.

Safety classification

Area	Position
Layer 1 through Layer 2.2	Physical workflow with preserved validation and regression evidence, subject to the stated input and grid contracts.
Layer 3.1A through 3.1D	Computationally verified research models. Not clinically calibrated or clinically validated.
Layer 3.2	Optional provisional non-local model. Hypothesis-generating only and disabled by default.
Clinical decisions	Outside the software claim. ASCEND does not approve plans or produce treatment recommendations.

Reading conventions

- GUI labels are shown in bold where they identify a page, tab, button, or stored status.
- Paths are examples. Replace them with approved local paths for the deployment environment.
- PASS, WARN, BLOCKED, STALE, NOT APPLICABLE, and OUTSIDE VALIDATED SCOPE have distinct meanings.
- Null or unavailable scientific values are not zero. The application preserves this distinction.

Sources used

The guide was derived from the executable package metadata, GUI source, application controller, CLI parser, example configuration, release records, scientific-kernel audit, DICOM and Eclipse contracts, Linux Docker sandbox report, cross-platform CI report, security policy, and current regression results.

CONTENTS

1. Software overview

ASCEND is a modular Lattice Radiotherapy analysis workstation. It imports a UID-linked radiotherapy DICOM chain, validates dose and structure geometry, calculates locked physical LRT metrics, constructs a spatial peak-to-valley graph, and optionally produces gated spatial radiobiology research outputs.

Interfaces	Native Qt workstation | CLI | localhost browser adapter
Workflow controller	Case state | configuration | invalidation | service dispatch | export
Research biology	Layer 3.1A LQ | 3.1B MLQ/EUD | 3.1C TR | 3.1D TCP | Layer 3.2 non-local
Validated physical analysis	Layer 2.1 LRT metrics | Layer 2.2 spatial PVDR graph
Validated input foundation	DICOM chain | Layer 1 dose, masks, geometry, DVH and provenance

Primary interfaces

Interface	Purpose	Scientific engine
Native Qt workstation	Interactive ten-page desktop workflow and 2D/3D evidence viewers.	Shared controller and services
Command line	Batch discovery, analysis, resumption, validation, cache, and export.	Shared controller and services
Localhost browser adapter	Optional local browser presentation layer.	Shared controller and services

Ten-page workflow

1 Import	2 Configure	3 Map roles	4 Layer 1	5 Layer 2.1
6 Layer 2.2	7 Layer 3.1	8 Layer 3.2	9 Review	10 Export

The workflow is deliberately ordered. Later layers consume stored, hash-linked upstream evidence. Running a downstream page does not authorize it to invent missing prescriptions, geometry, fraction history, tissue parameters, or model provenance.

Core design principles

- One selected DICOM chain prevents accidental mixing of dose, plan, structure, and image objects from different exports.
- RTSTRUCT SOP Instance UID plus ROI number is the authoritative structure identity; names are display and migration metadata.
- Scientific calculations are performed in services, never in GUI widgets or export renderers.
- Locked physical algorithms retain source-integrity hashes and are surrounded by auditable input, cache, status, and provenance services.
- Dependent results become STALE after relevant configuration changes instead of remaining deceptively current.

2. Supported environments and prerequisites

Validated operating-system matrix

Environment	Python	Current evidence	Scope
Ubuntu Linux x64	3.11 and 3.12	239 tests plus 28 subtests; clean install, GUI, 3D, numerical and package gates passed	Headless CI; Linux Docker adds interactive-capable X11 evidence
Windows x64	3.11 and 3.12	239 tests plus 28 subtests; clean install, GUI, 3D, numerical and package gates passed	Offscreen GUI and Mesa software OpenGL
Linux x64 minimum gate	3.9	Complete suite passed	Known pydicom 2.4.4 advisory limitation
macOS ARM64	3.11 and 3.12	Cross-platform matrix and package gates passed	Offscreen GUI and 3D

Preferred deployment baseline

Use 64-bit Python 3.11 or 3.12. These versions receive the clean dependency path used by current Windows and Linux CI. Python 3.9 remains installable but resolves an older pydicom version with a documented security advisory.

Required Python packages

Package	Declared range	Role
NumPy	>=1.24	Dose, masks, fields, metrics, and array integrity
pydicom	Version-marked by Python	DICOM discovery, decoding, relationships, and metadata
PySide6	>=6.7,<7	Native Qt workstation
PyVista	>=0.44,<1	VTK-backed 3D rendering and field visualisation
scikit-image	>=0.22,<1	Image and surface operations
SciPy	>=1.10,<2	Numerical algorithms and spatial analysis

Deployment prerequisites

- A local folder with sufficient capacity for DICOM inputs, native dose arrays, masks, derived fields, exports, logs, and temporary staging.
- Read and write permission for the case root. The source repository may remain read-only after installation.
- A supported OpenGL path for 3D use. CI proves software rendering; physical GPU and local driver behaviour require site acceptance testing.
- An approved PHI storage location. Do not use a public clone, synchronized personal folder, or Git working tree as a patient case root.

3. Installation

Clone the repository

```
git clone https://github.com/kaiwooodger/ASCEND.git
cd ASCEND
git checkout ci/github-hosted-phase2
```

Use the institutionally approved branch or release tag when one is designated. Record the commit identifier used for every retrospective analysis.

Windows PowerShell installation

```
py -3.12 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install -e .
python -m pip check
python run_ascend.py
```

If PowerShell blocks activation, apply the institution's approved script-execution policy or call the virtual-environment Python directly: `.venv\Scripts\python.exe run_ascend.py`.

Linux installation

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -e .
python -m pip check
python run_ascend.py
```

Typical Debian/Ubuntu Qt runtime libraries

```
sudo apt-get update
sudo apt-get install -y libgl1 libegl1 libxkbcommon-x11-0 libxcb-cursor0
```

Institutional deployment

Package installation, Git access, Python execution, and writable case storage may be controlled by RNSH policy. Use approved network, software, and PHI locations. Do not bypass endpoint controls.

Isolated verification

```
python -c "import ascend; print(ascend.__version__)"
python -m ascend.cli --help
python -m pip check
```

Expected package identity: 1.3.5. A mismatch indicates the wrong interpreter, environment, clone, or installed package.

4. Launching and first-run checks

Native workstation

```
python run_ascend.py
# equivalent after package installation
ascend gui
# equivalent module form
python -m ascend.cli gui
```

Optional browser adapter

```
python -m ascend.cli web-gui
```

> **Network boundary**
> The browser adapter is intended for localhost. Do not expose it beyond localhost without authentication, network architecture, threat modelling, and a separate security review.

Startup acceptance checks

1. Confirm the title bar displays ASCEND 1.3.5.
2. Confirm all ten workflow pages appear in the sidebar.
3. Confirm no Qt platform-plugin or OpenGL error appears in the launching terminal.
4. Open and close a file chooser to verify desktop integration.
5. Record the Git commit with `git rev-parse HEAD`.
6. Run a synthetic or approved non-clinical case before introducing clinical data.

Interface overview

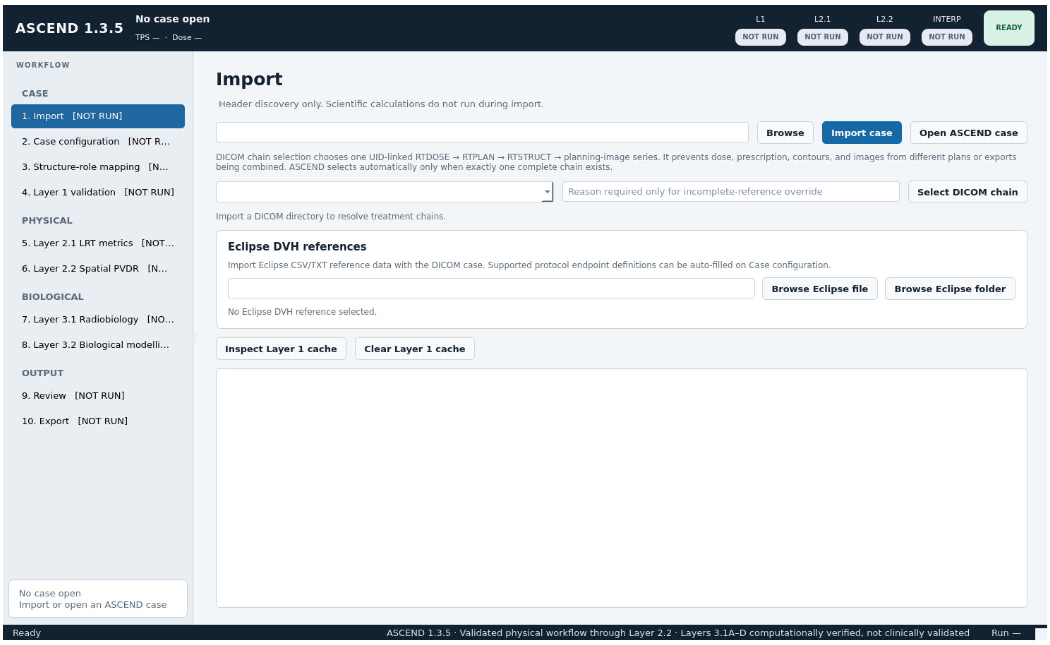

Figure 1. Native ASCEND workstation at startup. The sidebar enforces the ten-page workflow; the header and footer keep calculation and validation scope visible.

5. End-to-end operating workflow

1 Import	2 Configure	3 Map roles	4 Layer 1	5 Layer 2.1
6 Layer 2.2	7 Layer 3.1	8 Layer 3.2	9 Review	10 Export

Pre-analysis checklist

- A complete export directory containing RTDOSE, RTPLAN, RTSTRUCT, and the full referenced planning-image series.
- Known treatment approach, dose context, prescription values and sources, fractionation, treatment dates, and gaps.
- A structure-mapping plan for GTV, T_L, VTV_H, VTV_L, optional individual vertices, validation structures, and configured OARs.
- Eclipse DVH reference exports when independent TPS agreement is being assessed.
- Sourced biological parameters only when Layer 3 research analysis is required.

Operating rule

Do not skip unresolved gates

WARN can permit continuation when the warning is explicit and reviewable. BLOCKED, FAILED, STALE, and OUTSIDE VALIDATED SCOPE require resolution or documented non-applicability. Never reinterpret a missing value as zero.

Result hierarchy

For Layer 3.1, read outputs in this order: spatial map, whole-tumour mean surviving fraction and EUD, regional contribution, scenario/comparator result, TCP branch if configured, then provenance. This order prevents a summary number from being detached from its spatial and modelling assumptions.

What import does not do

- Import reads headers and constructs candidate DICOM chains.
- Import does not calculate dose metrics or radiobiology.
- Import does not infer prescription from RTDOSE maximum.
- Import does not decide that similarly named structures are identical.

6. Page 1 - Import

Input selection

1. Open Import and choose the DICOM export directory.
2. Optionally select an Eclipse CSV/TXT file or folder.
3. Press Import case. ASCEND performs header discovery only.
4. Review the candidate RTDOSE to RTPLAN to RTSTRUCT to image-series chains.
5. Accept automatic selection only when one complete chain exists. Otherwise choose the intended chain explicitly.

Incomplete-chain override

An incomplete chain can be selected only when patient identity and Frame of Reference are consistent and an explicit override reason is recorded. The override documents a known export omission; it does not make missing references valid.

Critical identity gate

Do not choose a chain by filename alone. Confirm plan label, UIDs, Frame of Reference, image series, dose object, and structure object.

Eclipse reference import

- Accepted forms: normalized ASCEND metrics CSV, Eclipse cumulative-DVH TXT, or a folder of TXT exports.
- Supported encodings include UTF-8, UTF-16, and Windows-1252.
- Supported dose units include Gy and cGy; supported volume units include cc, mL, and cubic centimetres.
- Patient, plan, normalization, and mapping ambiguity stop the reference import.

Layer 1 cache controls

Inspect Layer 1 cache lists case-local entries and integrity state. Clear Layer 1 cache requires confirmation and does not remove formal validated runs.

7. Page 2 - Case configuration

Treatment context

Configuration	Examples	Why it matters
Treatment approach	LRT alone; sequential LRT + cERT; integrated/SIB; unknown	Controls interpretation and fraction-event construction
Dose context	Selected plan or composite context	Prevents unsupported prescription interpretation
Rx_L and Rx_H	Explicit numeric values	Used only where endpoint definitions require them
Prescription provenance	RTPLAN, protocol, TPS, manual documented source	Prevents silent use of maximum dose as prescription
Treatment components	Component ID, source, fractions, dates, gaps	Builds biologically distinct fraction events

Protocol endpoints

Protocol endpoints are configured with structured role, endpoint type, and value controls. Supported roles include GTV, T_L, VTV_H, and VTV_L. Invalid roles, endpoint kinds, missing values, non-positive values, and D-percent values above 100 are rejected before calculation.

Configuration rules

- Enter prescriptions only from a documented source.
- Treat simultaneous, integrated, sequential, and composite courses as distinct contexts.
- Record fraction dates and gaps when time-dependent Layer 3 branches are requested.
- A missing prescription does not block physical endpoints that do not require it; the dependent normalized endpoint becomes not assessed.

No hidden defaults

ASCEND does not infer missing treatment meaning, prescription, fraction history, comparator schedule, or biological parameters from dose statistics.

8. Page 3 - Structure-role mapping

Authoritative identity

Each structure binding is stored as RTSTRUCT SOP Instance UID plus ROI number. Structure names remain visible for review but are not the calculation identity.

Role	Typical requirement	Purpose
GTV	Required	Gross tumour volume and primary tumour reporting domain
T_L	Configuration-dependent	Low-dose target role
VTV_H	Required for lattice metrics	Aggregate high-dose/vertex target
VTV_L	Required for peak-valley context	Valley or low-dose target role
VTV_H individual	Optional	Explicit per-vertex masks; preferred over connected-component derivation
Validation structures	Optional	Independent DVH and geometry checks
OARs	Optional	Identity-bound descriptive geometry and research field context

Rasterisation states

- rasterised: selected and successfully represented on the native RTDOSE grid.
- not_rasterised: intentionally unselected; no invented DVH, volume, or metric record.
- rasterisation_failed: selected but not processable; produces a blocking finding.

Optional OAR classification

Allowed classes are containing organ, target-excluded OAR, and separate critical OAR. The geometry module reports distances and overlaps only. It does not calculate protocol compliance or clinical pass/fail.

Mapping review

If another RTSTRUCT is selected, old identity bindings are invalidated. Re-map and rerun Layer 1 rather than retaining name-based assumptions.

9. Page 4 - Layer 1 validation

Purpose

Layer 1 converts the selected DICOM chain and mapped structures into a validated native-dose handoff. It enforces RTDOSE geometry, ROI identity, frame consistency, contour rasterisation, mask QA, TPS comparison inputs, cache integrity, and atomic publication.

Run sequence

1. Press Validate case.
2. Review DICOM-chain identity and geometry findings.
3. Review every required structure's rasterisation state, volume, dose, and provenance.
4. Review Eclipse agreement findings when a reference was supplied.
5. Confirm the Layer 2 eligibility gate.

Geometry contract

- Rows, columns, frames, position, orientation, spacing, frame offsets, scaling, units, and pixel dimensions are mandatory and strictly checked.
- Uniform anisotropic RTDOSE grids are accepted for Layer 1 and Layer 2.1.
- Non-uniform frame spacing is blocked.
- Frame offsets remain in source frame order; the application does not independently sort them.

Cache and publication

Layer 1 cache keys include input hashes, UIDs, chain selection, identity-bound structures, relevant configuration, reference hashes, and algorithm/schema versions. Formal runs are independently materialised, hash-verified, and atomically published.

Warning interpretation

A Layer 1 result may complete with warnings while retaining usable evidence. Read the warning text and downstream eligibility; do not convert WARN to PASS manually.

10. Page 5 - Layer 2.1 LRT metrics

Purpose

Layer 2.1 consumes the validated Layer 1 dose and masks and runs the locked six-metric physical LRT engine. Supporting-output services add QA, treatment context, geometry, protocol-endpoint status, and integrity provenance without changing the locked formulas.

Supporting evidence

- Per-vertex V95 relative to Rx_H, Dmean, D95, Dmax, and sampled volume.
- Explicit-vertex versus connected-component provenance and aggregate-mask consistency.
- High-dose coverage and volume-fraction context.
- Peak/valley sampled volumes, voxel counts, D50, mean-dose normalization, overlap, and outside-GTV warnings.
- Stored numerator and denominator for physical dose-ratio context.
- Layer 1 hashes, RTDOSE identity, native geometry, treatment context, warning inheritance, and applicability.

Explicit versus derived vertices

When individual VTV_H structures are mapped, ASCEND uses their validated RTSTRUCT masks. Otherwise, the locked algorithm derives deterministic connected components from aggregate VTV_H and labels them VTVH_CC_01, VTVH_CC_02, and so on. Provenance remains visible.

Prescription-dependent endpoints

Physical Dmean, D95, Dmax, and volume remain available without Rx_H. V95 relative to Rx_H becomes NOT ASSESSED rather than being calculated from an inferred prescription.

Operator review

1. Run Layer 2.1 or use the coordinated physical-analysis action.
2. Inspect the six primary metrics and their applicability.
3. Inspect per-vertex rows and source provenance.
4. Inspect supporting warnings before comparing cases.

11. Page 6 - Layer 2.2 Spatial PVDR

Purpose

Layer 2.2 constructs the stored vertex graph and local peak-to-valley dose-ratio evidence from the validated physical geometry. It preserves the locked nearest-neighbour, midpoint-sphere, and graph rules.

Validated grid scope

Scope gate

Layer 2.2 is validated for configured isotropic 1 mm and 2 mm native RTDOSE grids. Anisotropic grids or other isotropic spacings return OUTSIDE VALIDATED SCOPE while successful Layer 1 and Layer 2.1 results remain intact.

Viewer contents

- Transparent validated GTV envelope.
- Validated vertex surfaces in DICOM patient LPS millimetres.
- Stored graph edges coloured by local iPVDR.
- Locked 3 mm midpoint sampling spheres.
- Synchronized axial, sagittal, and coronal native-dose planes.
- Selected-edge evidence panel, warnings, components, and provenance.

Disconnected graph

A disconnected graph is a warning requiring geometric inspection. It is not automatically repaired or excluded because separation may reflect actual mapping, aggregate-mask derivation, or treatment geometry.

Mesh export

GTV and vertex surfaces can be exported as binary STL files in patient LPS millimetres with a JSON provenance manifest. The full graph export includes surfaces and stored edges as physical-scale connection tubes.

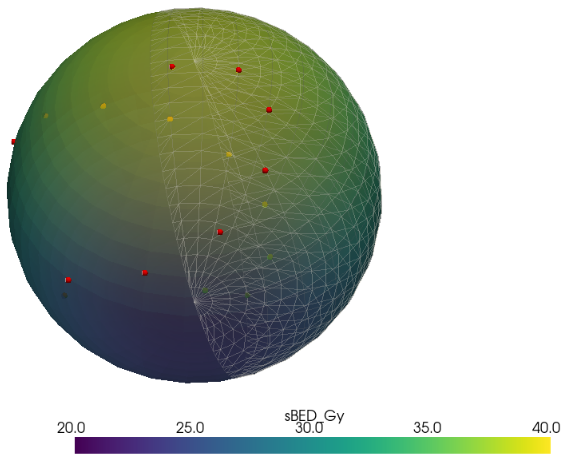

Figure 2. Linux sandbox VTK rendering evidence using software OpenGL. The synthetic surface demonstrates stored scalar mapping, geometry overlays, clipping, opacity, and colour-bar behaviour.

12. Page 7 - Layer 3.1 radiobiology

Research model

Layers 3.1A through 3.1D are computationally verified research models. They are not clinically calibrated, do not approve treatment plans, and must not be interpreted as patient-specific clinical outcome predictions.

Shared gates

- Validated upstream physical dose, masks, and geometry.
- Resolved and explicit fraction-event history.
- Identity-bound tissue parameter assignments with units and provenance.
- Hash-verified source arrays and matching spatial geometry.
- Branch-specific tumour, normal-tissue, comparator, time, or clonogen inputs.

Parallel branch structure

Branch	Output	Key boundary
3.1A	Voxelwise conventional LQ BED and EQD2	High-dose use remains a cautioned reference
3.1B	Guerrero-Li MLQ survival, effect, mean tumour SF, and EUD	Scenario parameters; not patient-specific radiosensitivity
3.1C	Modelled therapeutic ratio	Requires sourced normal parameters and a defined comparator schedule
3.1D	Direct-clonogenic Poisson TCP research branch	Biologically unvalidated; uniform density and model assumptions

3.1A equations

```
P(x) = sum_f d_f(x)
Q(x) = sum_f d_f(x)^2
BED(x) = P(x) + Q(x)/(alpha/beta)
EQD2(x) = BED(x)/(1 + 2/(alpha/beta))
```

Spatial transformation occurs before ROI reduction. A high-dose warning does not change the formula or switch the calculation to MLQ.

3.1B equations

```
z = lambda*tau + delta*d
G(z) = 2*(z + exp(-z) - 1)/z^2
K_f(x) = alpha*d_f(x) + beta*G(z_f(x))*d_f(x)^2
SF(x) = exp(-sum_f K_f(x))
```

Mean tumour survival is calculated over the validated GTV. EUD is obtained by bounded inversion under the explicit reference schedule. A separately configured normal-tissue field is required for OAR MLQ display or 3.1C.

13. Layer 3.1 configuration and guided sequence

Parameter configuration

- Assign alpha/beta values to identity-bound tumour and normal structures for LQ fields.
- Select or enter complete C1-C3 tumour kinetic parameter sets with source and parameter-set identifiers.
- Configure N1-N3 normal-tissue kinetic sets independently. Tumour kinetics are never substituted for normal tissue.
- Define a comparator schedule when 3.1C requires one. Sequential mixed-fraction courses do not receive an invented comparator.
- Configure clonogen density, units, source, and parameter-set identity for 3.1D.

Run and read order

1. Prepare the case and pass upstream gates.
2. Review treatment and physical dose history.
3. Select validated dose sources and confirm common geometry.
4. Assign tissue parameters.
5. Review conventional LQ high-dose caution.
6. Configure the tumour MLQ model.
7. Configure a normal model and comparator only when required.
8. Define therapeutic-ratio comparison requirements.
9. Save configuration and run Layer 3.1.

10-13. Inspect the spatial map and linked planes/CAD view.

14. Read whole-tumour mean surviving fraction and EUD.
15. Explain the result using regional contribution.
16. Assess scenarios, therapeutic ratio, and TCP where applicable.
17. Audit provenance and export.

Blocked branches

Independent branches can complete while another branch remains blocked. The overall run can therefore be completed with warnings. Read the branch state, reason, missing fields, numerical status, and provenance; do not infer a normal-looking value.

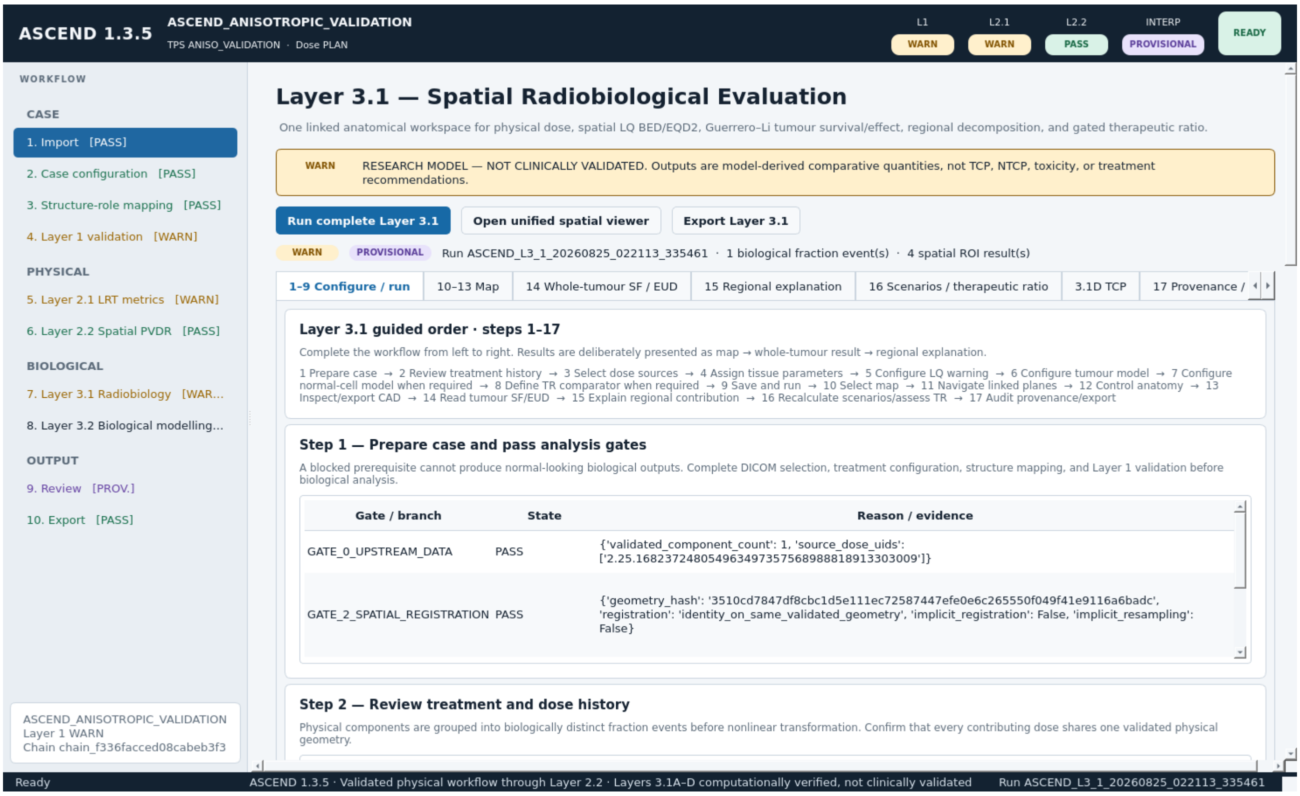

Figure 3. Layer 3.1 guided configuration/run page from the Linux end-to-end GUI harness. Gate evidence and the 17-step sequence are kept visible.

14. Layer 3.1 result interpretation

Map first

The spatial field is the primary result. Use linked axial, sagittal, and coronal views with one retained anatomy and crosshair. Complete-volume colour ranges are the default. Display transforms, clipping, opacity, and smoothing do not change stored science.

Whole-tumour result

For 3.1B, the primary numerical outputs are mean GTV surviving fraction and tumour EUD. Read their model, fraction schedule, parameter provenance, spatial mask, and solver status together.

Regional explanation

The regional cards and residual-survival contribution bar decompose stored GTV regions. Contributions sum to the whole-tumour residual-survival basis; the remainder region can dominate even when high-dose vertices show strong local effect.

Modelled therapeutic ratio

3.1C compares normal-tissue survival for the actual heterogeneous course with a normal-tissue survival reference at equal modelled tumour effect. It is calculated in log space and retains ln(TR) and log10(TR) when ordinary floating-point representation becomes unstable.

> **TR interpretation**
> Modelled therapeutic ratio is not clinical benefit, NTCP, toxicity, OAR compliance, or a plan-quality pass/fail score. A blocked comparator is not a ratio of zero.

3.1D direct-clonogenic Poisson TCP

```
mu_i = density * voxel_volume * SF_i * exp(repopulation_term)
N_surv = sum_i mu_i
TCP = exp(-N_surv)
ln(TCP) = -N_surv
```

- Consumes the authoritative 3.1B survival field and does not recalculate dose or MLQ survival.
- Reports radiation-only and optional repopulation-corrected results.
- Decomposes expected residual clonogens into vertex, valley, and remainder regions.
- Retains ln(TCP), log10(TCP), and numerical status when direct probability underflows to 0.0.

> **TCP interpretation**
> A displayed floating-point 0.0 can mean numerical underflow, not a failed kernel. Read expected surviving clonogens, ln(TCP), log10(TCP), numerical status, density provenance, and model assumptions.

15. Unified Layer 3.1 spatial viewer

Read-only contract

The viewer consumes hash-verified stored arrays. It never calculates BED, EQD2, MLQ survival, EUD, therapeutic ratio, TCP, dose resampling, masks, or geometry.

Available fields

- Physical absorbed dose.
- Spatial BED and EQD2 for the selected alpha/beta assignment.
- Tumour MLQ surviving fraction and effect, including display-only -log10(SF).
- Normal-tissue MLQ fields only when a separate complete normal parameter set was calculated.
- Residual-clonogen overlays for the configured 3.1D branch.

2D behaviour

- Shared anatomy, patient-LPS geometry, crosshair, and complete-volume range.
- Categorical masks are sampled as labels; quantitative fields retain their units.
- High-dose LQ warning regions remain outlines and do not replace field values.

3D/CAD behaviour

- Composite scene retains GTV, vertex, valley, and configured OAR surfaces independently.
- Volume rendering is the default; surface, isosurface, and slice modes are explicit alternatives.
- Anatomical or overlay smoothing is presentation-only and does not alter masks, arrays, metrics, or exports.
- BODY/EXTERNAL may be excluded from internal OAR display to prevent occlusion.

Paired-course comparison

A comparison is not inferred. The panel remains NOT CONFIGURED until a second hash-verified field set exists on the same validated geometry.

16. Page 8 - Layer 3.2 biological modelling

Disabled by default

Layer 3.2 is an optional provisional non-local research model. It is excluded from calculation and export unless explicitly enabled for the case.

What it models

Layer 3.2 consumes current Layer 3.1 evidence to calculate a dimensionless cumulative non-local mediator exposure H and an additional modelled survival reduction relative to baseline LQ. Physical absorbed dose is never modified.

Presentation

- Baseline-LQ survival, final survival, and additional-reduction panels.
- No-sink and available scenario comparisons; anatomical vascular sink remains unavailable without vessel geometry and an uptake model.
- Absolute consequence surfaces at 2.5%, 5%, 10%, and 20%.
- GTV, vertex, valley, peri-GTV shells, and configured-OAR regional summaries.
- Stored scalar VTI and geometry exports with versioned provenance.

Explicit exclusions

- No toxicity, clinical risk, NTCP, OAR compliance, or patient-outcome prediction.
- H is not physical dose and is not measured concentration.
- The no-vascular PDE does not imply a validated anatomical vascular model.
- CT pixels are not stored in the Layer 3.2 artifact.

Operating sequence

1. Complete and review Layer 3.1.
2. Explicitly enable Layer 3.2.
3. Select the documented preset and numerical parameters.
4. Run Layer 3.2 and inspect status, stored fields, regions, and provenance.
5. Build the hash-verified viewer only from the current completed result.

17. Pages 9 and 10 - Review and export

Review page

The Review page separates calculation status, interpretation status, applicability, warnings, errors, dependencies, and provenance. This separation prevents a numerically completed branch from being described as clinically interpretable.

Export principles

- JSON is authoritative.
- CSV files are presentation derivatives from stored result objects and do not recalculate science.
- Layer 3.1 provides its own research-result export pathway.
- Layer 3.2 evidence is omitted while the optional model is disabled.
- Array, geometry, source, parameter, and software hashes are retained where defined by the result contract.

Typical case exports

Artifact group	Examples
Canonical case result	ascend_result.json, ascend_case_config.json, ascend_summary.csv
Layer 2.1	metrics CSV, supporting-output JSON, per-vertex QA CSV
Layer 2.2	edge and node CSVs, viewer/mesh evidence when explicitly exported
Layer 3.1	radiobiological response JSON, LQ JSON, ROI metrics, response/TR/TCP tables, stored spatial fields
Layer 3.2	Versioned result JSON, NPZ/VTI fields, consequence geometry and provenance when enabled

Export check

1. Confirm the case and run IDs.
2. Confirm upstream and configuration hashes match the reviewed result.
3. Confirm blocked and not-applicable values remain null with reasons.
4. Store outputs only in the approved case location.
5. Do not add exports to Git.

18. Case storage, state, and invalidation

Case directory

```
case_root/
raw/ immutable input inventory and references
cache/layer1/ case-local hash-verified cache
validated/ Layer 1 dose, masks, geometry and findings
derived/layer2_1/ physical metric evidence
derived/layer2_2/ graph and spatial PVDR evidence
derived/layer3_1/ LQ, MLQ, TR, TCP and spatial fields
derived/layer3_2/ optional non-local research evidence
exports/ canonical JSON and CSV derivatives
logs/ text and structured JSONL events
ascend_case.json authoritative case state
```

Status meanings

Status	Meaning	Operator action
PASS / completed	Required calculation or gate completed.	Continue after reviewing provenance and warnings.
completed_with_warnings	Usable result with explicit caveats.	Review every warning before interpretation.
BLOCKED	Required input, identity, geometry, parameter, or dependency is missing/invalid.	Resolve the stated gate; do not infer a value.
STALE	A relevant upstream input or configuration changed.	Rerun the dependent layer.
OUTSIDE VALIDATED SCOPE	Execution domain is not covered by the locked validation contract.	Retain valid upstream outputs; do not claim validated downstream use.
NOT APPLICABLE	The branch does not apply to the configured context.	Preserve null and reason.
FAILED	Execution failed.	Read logs and error state; do not use partial output as final.

Selective invalidation

- Prescription changes do not rebuild Layer 1 but can stale prescription-dependent physical or biological outputs.
- RTSTRUCT, RTDOSE, image-series, selected-chain, or canonical-mapping changes invalidate Layer 1 and downstream results.
- Layer 3 parameter changes stale only dependent biological branches and viewers.
- Disabling Layer 3.2 invalidates or suppresses its evidence without mutating upstream results.

19. Command-line reference

Command	Purpose
ascend gui	Launch native workstation
ascend web-gui	Launch optional localhost browser adapter
ascend discover	Inspect DICOM headers and candidate chains without analysis
ascend run --case-root [options]	Import and run Layer 1 through Layer 2.2; optionally Layer 3.1; export
ascend resume --layer	Continue an existing case
ascend layer31 --case [--export]	Run gated Layer 3.1
ascend layer32 --case [--export]	Run enabled Layer 3.2
ascend cache-inspect	Inspect Layer 1 cache
ascend cache-clear --confirm	Clear case-local Layer 1 cache
ascend validate-eclipse-dvh ...	Compare stored ASCEND endpoints with reference Eclipse endpoints
ascend diagnose-eclipse-volumes ...	Investigate preserved volume discrepancies without changing formal results
ascend validate-anisotropic --output	Generate independent regular-anisotropic evidence
ascend validate-treatment-context --output	Generate treatment-context evidence

Batch example

```
python -m ascend.cli run /approved/dicom/export \
+ --case-root /approved/cases/CASE001 \
+ --config configs/ascend_case_config.example.json \
+ --with-layer31
```

Exit codes

Code	Meaning
0	Command completed within its contract
2	Selection, blocking, input, or execution failure
3	Physical workflow reached an outside-validated-scope result
4	Validation comparison completed with discrepancies or diagnostic findings

20. Eclipse DVH import and formal comparison

Reference import contract

- English Eclipse cumulative-DVH text or canonical CSV formats.
- Arbitrary patient, plan, and structure identifiers are supported when identity is unambiguous.
- One-structure-per-file and repeated-all-structures exports are supported.
- Direct dose is preferred; relative dose requires the exported total-dose normalization.
- Binary Eclipse objects, screenshots, PDFs, differential DVHs, localized labels, and unstable manually edited formats are rejected.

Formal comparison

The validation harness compares supplied Eclipse endpoints with endpoints already stored by ASCEND. It does not calculate new endpoints and does not mutate the validated pathway.

```
python -m ascend.cli validate-eclipse-dvh \
+ --case /approved/case/ascend_case.json \
+ --reference /approved/reference/eclipse.csv \
+ --output /approved/validation/eclipse \
+ --criteria configs/eclipse_dvh_acceptance_v1.json
```

Default software-agreement criteria

Endpoint type	Criterion
Dose	Absolute difference <= max(0.2 Gy, 2% of Eclipse dose)
Volume-at-dose percentage	Absolute difference <= 1.0 percentage point
Structure volume	Absolute difference <= max(0.1 cc, 2% of Eclipse volume)

Not a clinical tolerance

These are software-agreement criteria, not clinical treatment-plan constraints or protocol-compliance thresholds.

21. Data security and privacy

Protected data

DICOM files, case directories, logs, exports, screenshots, crash reports, derived arrays, masks, meshes, and even identifiers embedded in provenance can contain protected health information or reconstructable patient data.

Mandatory handling rules

- Keep patient data and case roots outside the Git repository.
- Do not commit DICOM, cases, validated runs, caches, exports, logs, screenshots, crash reports, arrays, or meshes.
- Use only explicitly synthetic, non-clinical fixtures in source control and automated tests.
- Inspect staged content before every push with `git status --short` and `git diff --cached`.
- If sensitive data enters Git history, deleting the latest file is insufficient. The history must be purged and exposed credentials rotated.
- Report vulnerabilities through a private security channel without patient data or exploitable detail in a public issue.

Recommended directory separation

Location	Contents
Source clone	Application code, synthetic tests, documentation; no clinical runtime data
Approved case storage	DICOM, case state, cache, validated outputs, exports, and logs
Approved validation storage	De-identified or authorized comparison evidence and reports

> **Repository boundary**
> A private GitHub repository is not automatically an approved clinical-data repository. Keep runtime patient evidence out of Git regardless of repository visibility.

22. Troubleshooting

Symptom	Likely cause	Resolution
No candidate chain	Missing RTDOSE/RTPLAN/RTSTRUCT/image references or unsupported export	Inspect discovery output; re-export a complete linked set
Chain selection required	Several complete chains exist	Review UIDs and plan context; select explicitly
Layer 1 blocked	Geometry, frame, pixel, identity, or rasterisation failure	Read the exact finding; correct source/export or mapping
Layer 2.2 outside scope	Anisotropic grid or spacing outside 1/2 mm contract	Retain valid Layer 1/2.1; do not claim validated Layer 2.2
Layer 3 fraction history unresolved	Fractions, components, dates, or dose-source mapping incomplete	Complete treatment context; rerun dependent branch
TR blocked	Normal kinetics or comparator missing	Provide complete sourced normal parameters and required schedule
TCP blocked	Density, units, source, parameter ID, mask, or survival dependency missing	Complete 3.1D provenance and rerun
TCP displays 0.0	Probability underflow with large residual clonogen burden	Read ln(TCP), log10(TCP), expected survivors, and numerical status
3D window blank or crashes	Qt/OpenGL/driver incompatibility	Verify runtime libraries; test software rendering; collect non-PHI logs
Result is STALE	Relevant input or configuration changed	Rerun the stated dependent layer
Wrong ASCEND version	Different Python environment or installed package	Check interpreter path and import location; reinstall in the intended environment

Diagnostic commands

```
python -c "import sys, ascend; print(sys.executable); print(ascend.__version__); print(ascend.__file__)"
python -m pip check
python -m ascend.cli discover /path/to/dicom
git rev-parse HEAD
```

Logs

Use the case logs and structured JSONL events to identify the failing gate, source identity, and dependency run. Remove or de-identify protected content before sharing diagnostics.

23. Validation evidence and limits

Current automated evidence

Evidence stream	Result
Current commit 65583e0	Cross-platform CI run 32820721676 passed
Ubuntu x64 Python 3.11/3.12	239 tests plus 28 subtests passed in each environment
Windows x64 Python 3.11/3.12	239 tests plus 28 subtests passed in each environment
Numerical equivalence	Cross-platform comparison gate passed
Clean package installation	Ubuntu, Windows, and macOS wheel validation passed
Linux Docker sandbox	Clean install, 216 tests plus 28 subtests, interactive-capable X11 GUI, DICOM E2E, software 3D, repeated sessions, and controlled error probes passed

What the evidence establishes

- Source, dependency, Python, numerical, DICOM, package, headless GUI, and headless 3D portability across the tested matrix.
- Interactive-capable Linux X11 operation under Xvfb/Openbox with Mesa llvmpipe.
- Software consistency of the Layer 3.1 kernels and presentation handoff.

What remains unvalidated

- Interactive Windows desktop behaviour on RNSH-managed endpoints.
- Physical GPU and site-specific driver behaviour.
- Windows ARM64, Ubuntu ARM64, and untested distribution/package combinations.
- Clinical deployment, workflow usability with clinical operators, patient-specific biological calibration, and treatment decisions.

Site acceptance
Successful CI and Docker results establish software portability, not automatic compatibility with every hospital security policy, GPU driver, storage topology, endpoint-control product, or clinical workflow.

24. Testing and maintenance

Developer environment

```
python -m pip install -e '.[test,quality]'
QT_QPA_PLATFORM=offscreen python -m pytest -q
python -m ruff check .
python -m compileall -q ascend benchmarks run_ascend.py tools
```

Required preservation

- Do not edit locked scientific sources without a formal validation and source-integrity process.
- Keep GUI, browser, and exporters as consumers of controller services and stored results.
- Add regression tests for every new DICOM export variant, geometry case, status transition, scientific gate, and output schema change.
- Record version, commit, dependency environment, synthetic fixtures, and acceptance criteria for validation runs.

Performance references

Profile	Uncached median	Cache-hit median	Peak RSS
Small synthetic	0.671 s	0.479 s	about 72 MB
Representative Eclipse	13.086 s	1.402 s	about 915 MB
Very large synthetic	72.989 s	2.946 s	about 2.06 GB

These are regression baselines, not guarantees for RNSH hardware. Run the benchmark suite on deployment hardware using approved non-clinical inputs before setting operational expectations.

Release control

- Use immutable Git tags or recorded commits for retrospective analyses.
- Require the aggregate cross-platform portability gate for release candidates.
- Preserve canonical exports and provenance rather than relying on screenshots or transcribed values.

25. Operator checklists

Installation acceptance

- [ ] Approved 64-bit Python 3.11 or 3.12 installed
- [ ] Repository cloned at recorded branch/tag/commit
- [ ] Virtual environment created and dependencies installed
- [ ] pip check passed
- [ ] ASCEND reports version 1.3.5
- [ ] Native GUI opens and all ten pages are present
- [ ] File dialog and 3D test work on the site endpoint
- [ ] Approved PHI case-storage location confirmed

Per-case analysis

- [ ] DICOM chain identity reviewed
- [ ] Treatment context and prescription provenance recorded
- [ ] Structure identities mapped by ROI number and RTSTRUCT UID
- [ ] Layer 1 findings and Layer 2 eligibility reviewed
- [ ] Layer 2.1 metrics and supporting outputs reviewed
- [ ] Layer 2.2 graph and scope state reviewed
- [ ] Layer 3 parameters are sourced and complete where used
- [ ] Maps, whole-tumour result, regional explanation, scenarios, and provenance read in order
- [ ] Export hashes, run IDs, blocked values, and warnings confirmed

Before sharing or publication

- [ ] No clinical runtime data staged in Git
- [ ] Exports and screenshots reviewed for identifiers
- [ ] Software version, commit, model parameters, and validation scope stated
- [ ] Research-model outputs are not described as clinical predictions

26. Glossary

Term	Meaning in ASCEND
ASCEND case	Persistent case root with authoritative ascend_case.json and linked evidence
DICOM chain	UID-linked RTDOSE, RTPLAN, RTSTRUCT, and planning-image series
GTV	Gross tumour volume reporting domain
VTV_H	High-dose/vertex target role
VTV_L	Valley or low-dose target role
LRT	Lattice Radiotherapy
PVDR / iPVDR	Peak-to-valley dose ratio / local spatial peak-to-valley evidence
BED	Biologically effective dose under the conventional LQ reference
EQD2	Equivalent dose in 2 Gy fractions under the conventional LQ reference
MLQ	Guerrero-Li modified linear-quadratic response model used by 3.1B
SF	Modelled surviving fraction
EUD	Survival-equivalent uniform dose under the configured MLQ schedule
TR	Modelled therapeutic ratio defined by the configured 3.1C comparator
TCP	Direct-clonogenic Poisson research output in 3.1D; not clinically calibrated
H	Dimensionless cumulative mediator exposure in optional Layer 3.2
LPS	DICOM patient coordinate convention: left, posterior, superior
STALE	Stored output no longer current after a relevant input/configuration change
OUTSIDE VALIDATED SCOPE	Input is processable upstream but not covered by the downstream validation contract

Document references

Repository: github.com/kaiwooodger/ASCEND

Current cross-platform run: GitHub Actions run 32820721676

Key repository records: README.md; SECURITY.md; docs/RELEASE_1.3.5.md; docs/DICOM_INGESTION_V2.md; docs/LAYER21_SUPPORTING_OUTPUTS.md; docs/LAYER31_LQ_BED_EQD2.md; docs/LAYER31_SCIENTIFIC_KERNEL_AUDIT_2026-08-25.md; docs/LAYER31_SPATIAL_BIOLOGY_VIEWER.md; docs/LAYER32_NONLOCAL_EFFECT.md; validation/linux/LINUX_SANDBOX_VALIDATION.md; validation/cross_platform/GITHUB_HOSTED_VALIDATION.md.

End of guide

This document describes the current repository baseline. Reissue it when software identity, workflow pages, model contracts, validation scope, dependencies, or deployment evidence changes.